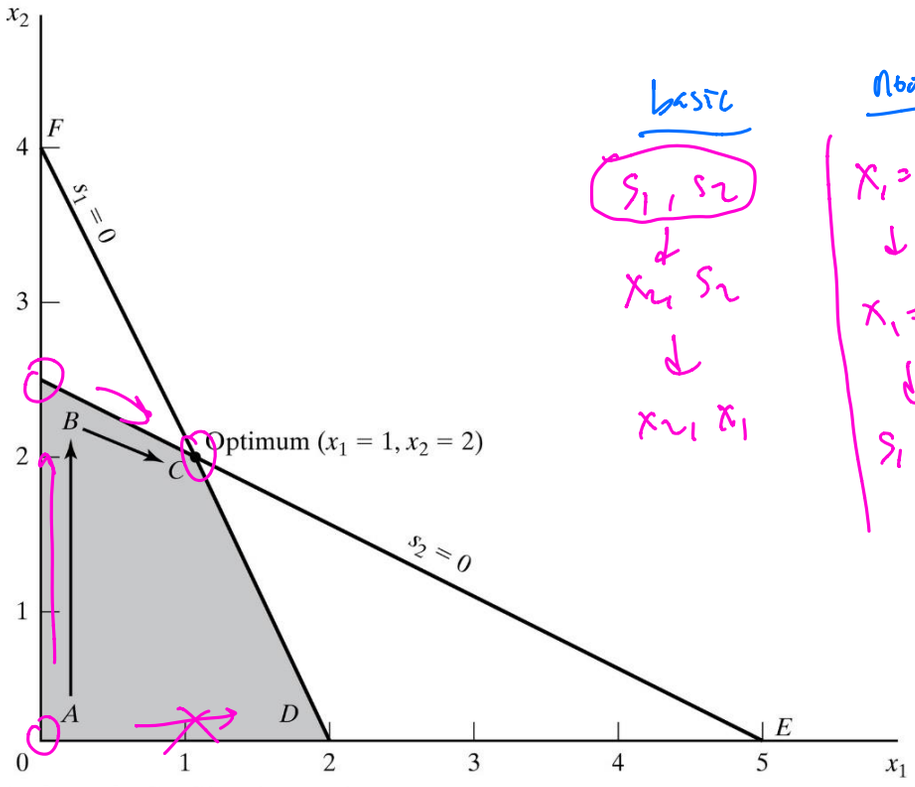
Iterative Nature of the Simplex Method

- Simplex method usually starts at origin.
- At each iteration 1 basic variable changes
- The non-basic variable which improves the objective function most is selected to be basic.



A → B → C : Simplex path

At A $x_1 = x_2 = 0$

max $z = 2x_1 + 3x_2$

x_2 has a larger coefficient

→ move to B

At point B, the simplex method will increase the value of x_1 to reach the corner point C, the optimum.

The path of simplex algorithm always connects corner points.

It always moves alongside the edges of the solution space, i.e. it does not cut across the solution space.

The Simplex Method

Remember the Reddy Mikks example:

Max $z = 5x_1 + 4x_2$

s.t.

$$\begin{aligned} 6x_1 + 4x_2 &<= 24 \quad (M1) \\ x_1 + 2x_2 &<= 6 \quad (M2) \\ -x_1 + x_2 &<= 1 \quad (demand) \\ x_2 &<= 2 \quad (demand) \end{aligned}$$

→

$z - 5x_1 - 4x_2 - 0s_1 - 0s_2 - 0s_3 - 0s_4 = 0$

Max. $z = 5x_1 + 4x_2 + 0s_1 + 0s_2 + 0s_3 + 0s_4$

s.t

$$\begin{aligned} 6x_1 + 4x_2 + s_1 &= 24 \\ x_1 + 2x_2 + s_2 &= 6 \\ -x_1 + x_2 + s_3 &= 1 \\ x_2 + s_4 &= 2 \end{aligned}$$

Simplex Tableau (pivot row, entering variable)

Basic	z	x_1	x_2	s_1	s_2	s_3	s_4	solution
z	1	-5	-4	0	0	0	0	0
s_1	0	6	4	1	0	0	0	24
s_2	0	1	2	0	1	0	0	6
s_3	0	-1	1	0	0	1	0	1
s_4	0	0	1	0	0	0	1	2

leaving variable

n-m → 6-4 → non-basic → 0

M ← basic → non-zero → 4

pivot row

The algorithm starts at the origin, i.e. $(x_1, x_2)=(0,0)$ ✓

- Non-basic → (x_1, x_2) ✓
- Basic → (s_1, s_2, s_3, s_4) ✓

$z = 0$
$s_1 = 24$
$s_2 = 6$
$s_3 = 1$
$s_4 = 2$

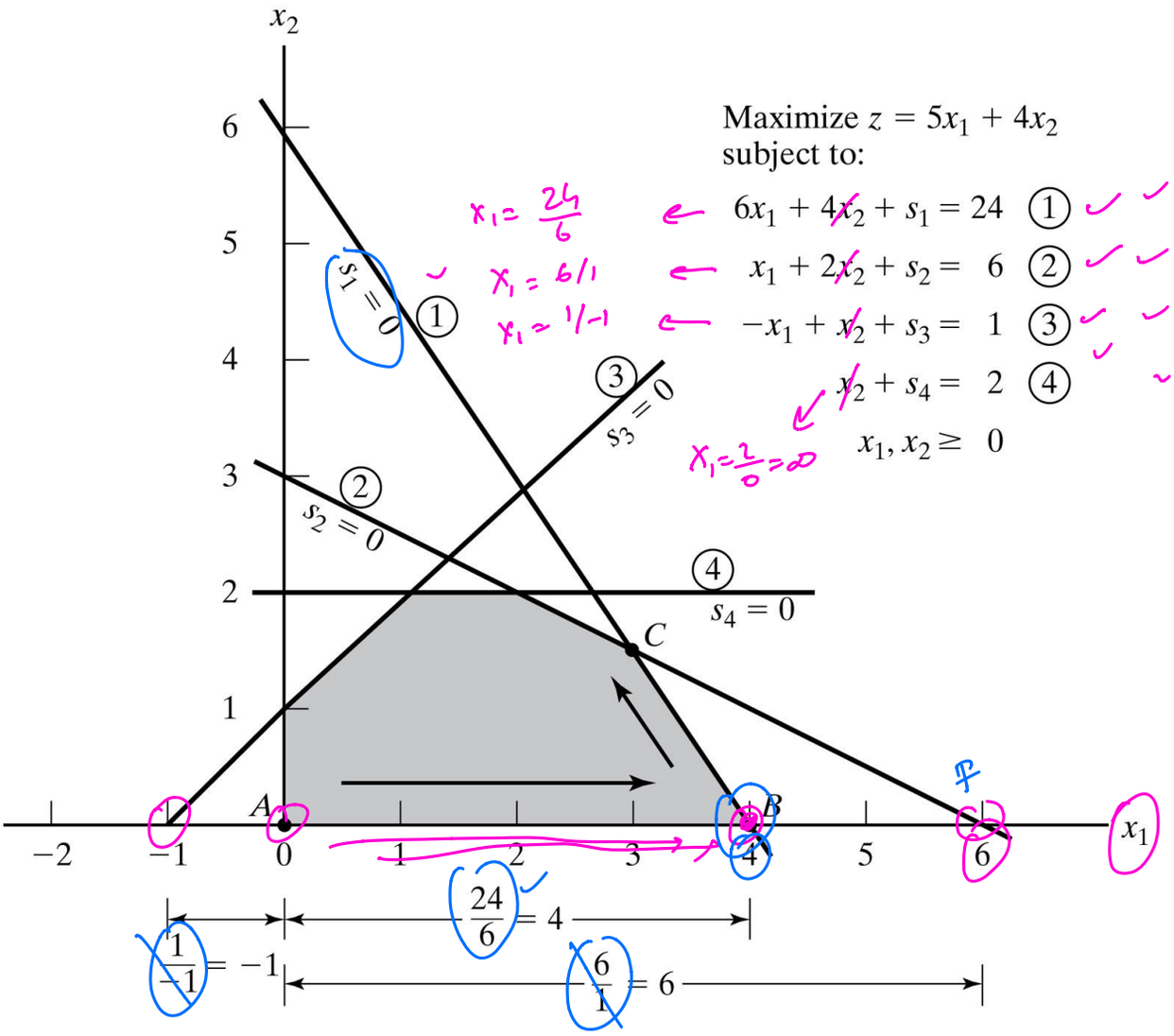
} Due to the identity matrix

$z = 5x_1 + 4x_2$ or $z - 5x_1 - 4x_2 = 0$ → x_1 should enter the basis (entering variable)

basic set

Most positive — Most negative

- The z equation shows that the solution can be improved by improving the value of x_1 or x_2 above zero. x_1 is selected because it has the most positive objective coefficient.
- This rule defines the so-called **simplex optimality condition**.
- If one is entering the basic variable set, one must leave (because # of non-basic variable must always be n-m)



Maximize $z = 5x_1 + 4x_2$
subject to:

$$6x_1 + 4x_2 + s_1 = 24 \quad (1)$$
$$x_1 + 2x_2 + s_2 = 6 \quad (2)$$
$$-x_1 + x_2 + s_3 = 1 \quad (3)$$
$$x_2 + s_4 = 2 \quad (4)$$
$$x_1, x_2 \geq 0$$

To determine the leaving basic variable, apply minimum ratio (MR) test: (this assures feasibility)

minimum positive ratio.

Basic	x_1	Solution	Ratio	
s_1	6	24	24/6 = 4	
s_2	1	6	6/1 = 6	
s_3	-1	1	1/-1 = -1	(ignore)
s_4	0	2	2/0 = ∞	(ignore)

simplex feasibility condition

when x_1 is entering the basis, we compute the intercepts of all the constraints with the non-negative direction of x_1-axis: These intercepts are the ratios.

MR appeared in s_1-row
→ s_1 is the leaving variable
→ x_1=4 (the MR) (point B)

non-basic → (s_1, x_2)
basic → (x_1, s_2, s_3, s_4) } Point B

To obtain the new tableau apply the Gauss-Jordan row operations:

1. Pivot row npr = cpr / pe
   New pivot row = current pivot row / pivot element
2. All other rows including z
   New row = current row – (its pivot col. Coef.) x (new pivot row)

   cr npr

New pivot row = current row / 6
New z-row = current z-row – (-5) x new pivot row
New s_2-row = current s_2-row – (1) x new pivot row
New s_3-row = current s_3-row – (-1) x new pivot row
New s_4-row = current s_4-row – (0) x new pivot row

pivot element

current z-row (czr)

Basic	z	x_1	x_2	s_1	s_2	s_3	s_4	solution	
z	1	-5	-4	0	0	0	0	0	
s_1	0	6	4	1	0	0	0	24	
s_2	0	1	2	0	1	0	0	6	← csr
s_3	0	-1	1	0	0	1	0	1	
s_4	0	0	1	0	0	0	1	2	
		x_1	x_2	s_1	s_2	s_3	s_4	s_{-1}	
entering variable									
z	1	0	-4/3	5/6	0	0	0	20	
pivot element									
x_1	0	1	2/3	1/6	0	0	0	4	← pr
leaving ← s_2	0	0	4/3	-1/6	1	0	0	2	
s_3	0	0	5/3	1/6	0	1	0	5	
s_4	0	0	1	0	0	0	1	2	

$z = \frac{2}{3}x_2 - \frac{5}{6}s_1 + 20$ Increasing x_2 is beneficial →

Basic	x_2	Sol.	Ratio
x_1	2/3	4	4/(2/3) = 6
s_2	4/3	2	2/(4/3) = 1.5
s_3	5/3	5	5/(5/3) = 3
s_4	1	2	2/1 = 2

New pivot row = current row / 1.5
New z-row = current z-row – (-2/3) x new pivot row
New x_1-row = current x_1-row – (2/3) x new pivot row
New s_3-row = current s_3-row – (5/3) x new pivot row
New s_4-row = current s_4-row – (1) x new pivot row

Basic	z	x_1	x_2	s_1	s_2	s_3	s_4	sol.
z	1	0	0	3/4	1/2	0	0	21
x_1	0	1	0	1/4	−1/2	0	0	3
x_2	0	0	1	−1/8	3/4	0	0	3/2
s_3	0	0	0	3/8	−5/4	1	0	5/2
s_4	0	0	0	1/8	−3/4	0	1	1/2

$z = -\frac{3}{4}s_1 + \frac{1}{2}s_2$

Since z-row coefficient of all non-basic variables are non-negative, the current tableau is optimal with:

$x_1 = 3$ ✓
$x_2 = 3/2$ ✓
$s_3 = 5/2$ ✓
$s_4 = 1/2$ ✓

$z = 21 = 5x_1 + 4x_2 = 5 \times 3 + 4 \times \frac{3}{2} = 21$

$s_1=0$ → M_1 is used completely → M_1 is scarce
$s_2=0$ → M_2 is used completely → M_2 is scarce

if $s_i > 0$ → M_i is abundant

In a minimization problem everything is the same except that the entering variable is determined as the one having most positive coefficient in z-row.

Simplex Method Summary

Optimality Condition: In a max (min) problem the non-basic variable having the most negative (positive) coefficient in the z-row is the entering variable.

Feasibility Condition: For both max and min problems the basic variable having the smallest non-negative ratio is the leaving variable.

Step 0. Determine a starting basic feasible solution.
Step 1. Select the entering variable using the optimality condition.
Stop if there is no entering variable; last solution is optimal.
Step 2. Select a leaving variable using the feasibility condition.
Step 3. Determine the new basic solution by Gauss-Jordan row operations. Go to step 1.

Example:

max. $2x_1 + x_2 - 3x_3 + 5x_4 \rightarrow 41$ (6, 7)

s.t
$x_1 + 2x_2 + 2x_3 + 4x_4 \le 40$
$2x_1 - x_2 + x_3 + 2x_4 \le 8$
$4x_1 - 2x_2 + x_3 - x_4 \le 10$

---

max. $2x_1 + x_2 - 3x_3 + 5x_4 + 0x_5 + 0x_6 + 0x_7$

s.t
$x_1 + 2x_2 + 2x_3 + 4x_4 + x_5 = 40$
$2x_1 - x_2 + x_3 + 2x_4 + x_6 = 8$
$4x_1 - 2x_2 + x_3 - x_4 + x_7 = 10$

---

entering var.

Basic	z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	sol.	MR
z	1	-2	-1	3	-5	0	0	0	0	
x_5	0	1	2	2	4	1	0	0	40	40/4 = 10
x_6	0	2	-1	1	2	0	1	0	8	8/2 = 4 ← pr
x_7	0	4	-2	1	-1	0	0	1	10	10/-1 = -10

← leaving var.

---

entering var.

Basic	z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	sol	MR
z	1	3	-7/2	11/2	0	0	5/2	0	20	← z + 5.npr
← leaving var. x_5	0	-3	4	0	0	1	-2	0	24	24/4 = 6 ← x_5 − 4.npr
x_4	0	1	-1/2	1/2	1	0	1/2	0	4	4/-1/2 = -8 ← npr = pr/2
x_7	0	5	-5/2	3/2	0	0	1/2	1	14	14/-1/2 = -28/5 ← x_7 + npr

Basic	z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	sol
z	1	3/8	0	11/2	0	7/8	7/4	0	41 ← z + 7/2.npr
x_2	0	-3/4	1	0	0	1/4	-1/2	0	6 ← npr = pr/4
x_4	0	5/8	0	1/2	1	1/8	1/4	0	7 ← x_4 + 1/2 npr
x_7	0	25/8	0	3/2	0	5/8	-3/4	1	29 ← x_7 + 5/2 npr

opt → $x_1 = 0$, $x_2 = 6$, $x_3 = 0$, $x_4 = 7$ ⇒ $z = 41$

max $2x_1 + x_2 - 3x_3 + 5x_4 = 41$

Example: min. $5x_1 - 4x_2 + 6x_3 - 8x_4$

s.t $x_1 + 2x_2 + 2x_3 + 4x_4 + x_5 = 40$

$2x_1 - x_2 + x_3 + 2x_4 + x_6 = 8$

$4x_1 - 2x_2 + x_3 - x_4 + x_7 = 10$

entering var. x_4

Basic	z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	sol.	MR
z	1	-5	4	-6	8	0	0	0	0	
x_5	0	1	2	2	4	1	0	0	40	40/4 = 10
x_6	0	2	-1	1	2	0	1	0	8	8/2 = 4
x_7	0	4	-2	1	-1	0	0	1	10	10/-1 = -10

leaving var. ← x_6

entering var. x_2

Basic	z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	sol.	
z	1	-13	8	-10	0	0	-4	0	-32	MR ← z − 8·npr
x_5	0	-3	4	0	0	1	-2	0	24	24/4 = 6 ← x5R − 4·npr
x_4	0	1	-1/2	1/2	1	0	1/2	0	4	4/-1/2 = -8 ← npr = pr/2
x_7	0	5	-5/2	3/2	0	0	1/2	1	14	14/-5/2 = 28/5 ← x7R

leaving var. ← x_5

Basic	z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	sol.	
z	1	-7	0	-10	0	-2	0	0	-80	← z − 8·npr
x_2	0	-3/4	1	0	0	1/4	-1/2	0	6	← npr = pr/4
x_4	0	5/8	0	1/2	1	1/8	1/4	0	7	← x4R + 1/2 npr
x_7	0	21/8	0	3/2	0	5/8	-1/4	1	29	← x7R + 5/2 npr

min $5x_1 - 4x_2 + 6x_3 - 8x_4 = z = -80$

$x_1 = 0$, $x_2 = 6$, $x_3 = 0$, $x_4 = 7$